

Code Evolution for Control: Synthesizing Policies via LLM-Driven Evolutionary Search

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Abstract—Designing effective control policies for autonomous systems remains a fundamental challenge, traditionally addressed through reinforcement learning or manual engineering. While reinforcement learning has achieved remarkable success, it often suffers from high sample complexity, reward shaping difficulties, and produces opaque neural network policies that are hard to interpret or verify. Manual design, on the other hand, requires substantial domain expertise and struggles to scale across diverse tasks. In this work, we demonstrate that LLM-driven evolutionary search can effectively synthesize interpretable control policies in the form of executable code. By treating policy synthesis as a code evolution problem, we harness the LLM’s prior knowledge of programming patterns and control heuristics while employing evolutionary search to explore the solution space systematically. We implement our approach using EvoToolkit, a framework that seamlessly integrates LLM-driven evolution with customizable fitness evaluation. Our method iteratively evolves populations of candidate policy programs, evaluating them against task-specific objectives and selecting superior individuals for reproduction. This process yields compact, human-readable control policies that can be directly inspected, modified, and formally verified. Experimental results on the LunarLander benchmark demonstrate that our approach achieves a 70% landing success rate, surpassing the 60% of PPO, while producing compact 59-line policies that can be directly inspected and verified. Although the average reward is lower than neural network policies, the evolved code exhibits superior interpretability and reliability in safety-critical metrics. This work highlights the potential of combining foundation models with evolutionary computation for synthesizing trustworthy control policies in autonomous systems. Code is available at <https://github.com/pgg3/EvoControl>.

Index Terms—LLM, evolutionary computation, control policy, code synthesis, interpretability

I. INTRODUCTION

Designing effective control policies for autonomous systems remains a fundamental challenge in robotics and artificial intelligence. Traditional approaches fall into two categories: reinforcement learning (RL) and manual engineering. While RL methods like Proximal Policy Optimization (PPO) [1] have achieved remarkable success, they suffer from high sample complexity, reward shaping difficulties, and produce opaque neural network policies that resist interpretation and verification [2]. Manual design, conversely, requires substantial domain expertise and struggles to scale across diverse tasks.

For safety-critical applications, the inherent opacity of neural network policies presents a fundamental obstacle. When

failures occur, diagnosing the root cause is often intractable. This precludes formal verification, complicates systematic debugging, and ultimately undermines trust in deployed autonomous systems.

To address these limitations of neural network policies, recent research has explored *programmatic policies*, which express control logic as executable code [3], [4]. Such policies provide inherent transparency: they can be directly inspected, modified, and formally verified. However, synthesizing effective programmatic policies remains difficult. Manual implementation demands significant domain expertise, while automated approaches face their own obstacles: designing domain-specific languages requires substantial upfront effort, and traditional program search methods like genetic programming are prone to combinatorial explosion [5].

LLM-driven code evolution has recently achieved notable success across diverse domains. FunSearch [6] discovered novel mathematical constructions by evolving program components, while CoEvo [7] extended this paradigm to symbolic regression with a dynamic knowledge library for open-ended discovery. In the optimization domain, Evolution of Heuristics [8] applied LLM-based mutation and crossover to automatically design algorithms, and EvoEngineer [9] optimized CUDA kernels through reflection-guided code evolution. These methods demonstrate that LLMs can serve as intelligent search operators, generating semantically meaningful variations that traditional approaches struggle to produce. However, their application to control policy synthesis remains unexplored.

This observation motivates the following research questions:

- **RQ1:** Can LLM-driven evolution synthesize control policies that match or exceed RL baselines?
- **RQ2:** How do different LLM evolution strategies compare for policy synthesis?
- **RQ3:** Do evolved code policies offer interpretability advantages over neural networks?

To address these questions, we propose treating policy synthesis as a code evolution problem. The LLM serves dual roles: as a mutation operator that generates semantically meaningful code variations, and as a crossover operator that recombines promising components across individuals. This approach harnesses the LLM’s prior knowledge of programming

patterns and control heuristics while using evolutionary search to systematically explore the solution space.

Our contributions are as follows:

- We formalize control policy synthesis as a code evolution problem and demonstrate its feasibility using LLM-driven evolutionary search.
- We systematically compare three LLM evolution strategies (FunSearch, EoH, and EvoEngineer) on the LunarLander control benchmark.
- We show that evolved policies achieve competitive performance with significantly better interpretability than neural network alternatives.

II. RELATED WORK

A. Reinforcement Learning for Control

Reinforcement learning has become the dominant paradigm for learning control policies in robotics and autonomous systems. Value-based methods like DQN [10] learn action-value functions, while policy gradient methods like PPO [1] directly optimize parameterized policies. Despite impressive results, RL methods typically require millions of environment interactions and produce opaque neural network policies that resist interpretation [2].

B. Program Synthesis and Neuro-Symbolic Methods

Program synthesis aims to automatically generate programs from specifications [11]. Recent neuro-symbolic approaches combine neural networks with symbolic reasoning to improve interpretability [12]. However, these methods often require domain-specific languages or extensive supervision.

C. LLM-Driven Code Evolution

Large language models have demonstrated remarkable code generation capabilities [13]–[15], and recent work has begun exploring their use to iteratively improve code [16], [17].

A particularly promising direction combines LLMs with evolutionary computation. FunSearch [6] pioneered this approach, using LLMs to evolve programs that discovered new mathematical constructions with improved bounds for the cap set problem. CoEvo [7] extended this paradigm to symbolic regression, introducing a dynamic knowledge library that enables continual, open-ended discovery. Evolution of Heuristics (EoH) [8] employed LLMs as mutation and crossover operators for combinatorial optimization, demonstrating that semantic code transformations significantly outperform random perturbations. EvoEngineer [9] applied reflection-guided LLM evolution to CUDA kernel optimization, achieving substantial speedups over baseline implementations.

Our work extends this line of research to control policy synthesis, where the goal is to evolve interpretable Python functions that map states to actions.

Algorithm 1 LLM-Driven Policy Evolution

Require: Environment $\mathcal{E}$, population size n , generations G

Ensure: Best policy π^*

```

1: Initialize population  $P \leftarrow \{\pi_1, \dots, \pi_n\}$  via LLM
2: for  $g = 1$  to  $G$  do
3:   for each  $\pi_i \in P$  do
4:      $f(\pi_i) \leftarrow \text{EVALUATE}(\pi_i, \mathcal{E})$ 
5:   end for
6:    $P_{\text{parents}} \leftarrow \text{SELECT}(P, f)$ 
7:    $P_{\text{offspring}} \leftarrow \text{LLM-EVOLVE}(P_{\text{parents}})$ 
8:    $P \leftarrow \text{REPLACE}(P, P_{\text{offspring}}, f)$ 
9: end for
10: return  $\arg \max_{\pi \in P} f(\pi)$ 

```

III. METHODOLOGY

A. Problem Formulation

We formalize control policy synthesis as a code evolution problem. Given an environment $\mathcal{E}$ with state space $\mathcal{S}$, action space $\mathcal{A}$, and reward function $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, we seek to find a policy function $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximizes the expected cumulative reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T R(s_t, \pi(s_t)) \right] \quad (1)$$

Unlike neural network policies, we constrain π to be an interpretable Python function that can be directly inspected, modified, and verified.

B. LLM-Driven Evolution Framework

Our approach maintains a population $P = \{\pi_1, \pi_2, \dots, \pi_n\}$ of candidate policy programs and evolves them using LLM-based operators. The overall procedure is summarized in Algorithm 1.

The key innovation lies in LLM-EVOLVE (line 7), which uses LLMs to generate semantically meaningful code transformations. We compare three strategies for this operator:

FunSearch [6] uses a single-prompt approach, showing the LLM a progression from worse to better solutions and asking it to continue the improvement trajectory.

EoH [8] employs five specialized operators (initialization, exploration, guided crossover, structural mutation, and parameter mutation), enabling both exploration and exploitation of the solution space.

EvoEngineer [9] provides rich context including episode statistics, failure modes, and domain-specific hints, requiring structured responses with code and rationale.

C. Fitness Evaluation

For control tasks, fitness is computed as the average reward over multiple episodes:

$$f(\pi) = \frac{1}{K} \sum_{k=1}^K \sum_{t=0}^{T_k} R(s_t^{(k)}, \pi(s_t^{(k)})) \quad (2)$$

TABLE I
PERFORMANCE COMPARISON ON LUNARLANDER-V3. RESULTS FOR LLM
METHODS ARE AVERAGED OVER 5 SEEDS.

Method	Avg Reward	Success%	LLM Calls	LoC
Random	-200 ± 50	0%	-	N/A
PPO (1M steps)	214 ± 27	60%	-	N/A
FunSearch	9.1 ± 33.8	22%	~ 45	20-35
EoH	6.2 ± 51.2	16%	~ 45	25-40
EvoEngineer	66.6 ± 56.1	40%	~ 45	30-50
EvoEngineer+	143.6	70%	~ 200	59

where K is the number of evaluation episodes and T_k is the episode length.

Policies are executed in a sandboxed environment with restricted builtins to ensure safety. Invalid actions or runtime errors result in minimum fitness scores.

IV. EXPERIMENTS

A. Experimental Setup

Task Environment. We evaluate on LunarLander-v3 from Gymnasium [18], a classical control benchmark where the agent must land a spacecraft on a designated pad. The environment provides an 8-dimensional continuous state (position, velocity, angle, angular velocity, leg contacts) and 4 discrete actions (do nothing, fire left/main/right engine). The reward function incentivizes precise landing (+100 to +140) while penalizing crashes (-100) and fuel consumption. An episode achieving cumulative reward ≥ 200 is considered successful.

Baselines. We compare three LLM-driven evolution strategies against two baselines: (1) *Random*, which selects actions uniformly at random; (2) *PPO* [1], trained for 1M timesteps using Stable-Baselines3 [19] with default hyperparameters. The three LLM-based methods (FunSearch, EoH, and EvoEngineer) are configured identically for fair comparison.

Configuration. All LLM evolution experiments use GPT-4o as the base model with temperature 0.7. We maintain a population of 10 candidate policies and evolve for 20 generations. Each policy’s fitness is computed as the average reward over 10 evaluation episodes. Results are averaged over 5 independent runs with different random seeds.

Metrics. We assess methods along three dimensions: (1) *Performance*: average episode reward and success rate (reward > 200); (2) *Efficiency*: number of LLM API calls and environment episodes consumed; (3) *Interpretability*: lines of code and cyclomatic complexity of the final policy.

V. RESULTS

A. Main Results

Table I summarizes the performance of all methods on LunarLander-v3.

Comparison among LLM-based methods. Among the three evolution strategies, EvoEngineer significantly outperforms both FunSearch and EoH. With an identical budget of ~ 45 LLM calls, EvoEngineer achieves an average reward of 66.6 and 40% success rate, while FunSearch and EoH remain near zero (9.1 and 6.2 respectively). This $7\times$ improvement

TABLE II
COMPARISON AT EQUIVALENT ENVIRONMENT INTERACTIONS (~ 1 M
STEPS).

Method	Env Steps	Reward	Success%
PPO	1M	214	60%
EvoEngineer+	~ 1 M	144	70%

in average reward demonstrates that providing rich context (episode statistics, failure modes, domain hints) is crucial for effective policy evolution. The structured prompts in EvoEngineer enable the LLM to make more informed code modifications, whereas simpler approaches struggle to discover effective control strategies.

Comparison with PPO. When extended to 200 LLM calls (EvoEngineer+), consuming approximately 1M environment steps comparable to PPO’s training budget, our approach achieves 143.6 average reward. More importantly, EvoEngineer+ attains a **70% success rate, exceeding PPO’s 60%**. While PPO achieves higher average reward due to more consistent performance across episodes, the evolved policy demonstrates superior landing success.

The interpretability advantage. The key distinction is that EvoEngineer+ produces a 59-line Python function that can be directly inspected, understood, and modified by human engineers. In contrast, PPO’s policy is encoded in ~ 10 K neural network parameters that offer no insight into the decision-making logic. This interpretability comes at modest performance cost (67% of PPO’s average reward) while achieving higher task success rate, representing an attractive trade-off for safety-critical applications where understanding policy behavior is essential.

B. Convergence Analysis

Figure 1 shows the convergence curves of EoH and EvoEngineer.¹ EvoEngineer demonstrates faster initial convergence, reaching positive rewards within the first few generations while EoH struggles to escape near-zero performance early on. Both methods show continued improvement throughout the evolution process, indicating that LLM-driven search can progressively refine policy quality. The shaded regions reveal high variance across seeds, reflecting the stochastic nature of LLM-based code generation. Different random seeds lead to different evolutionary trajectories, though EvoEngineer consistently outperforms EoH across all runs.

C. Sample Efficiency Analysis

To compare LLM-driven evolution with RL under equivalent environment interaction budgets, we conducted an experiment where EvoEngineer runs for 200 LLM calls (50 generations), consuming approximately 1M environment steps to match PPO’s budget. Table II summarizes this comparison.

At equivalent sample budgets, EvoEngineer+ achieves a 70% success rate compared to PPO’s 60%, though PPO attains

¹EoH runs for 10 generations while EvoEngineer runs for 12, as EoH uses 5 operators per generation versus EvoEngineer’s 3, resulting in different generation counts for the same LLM call budget.

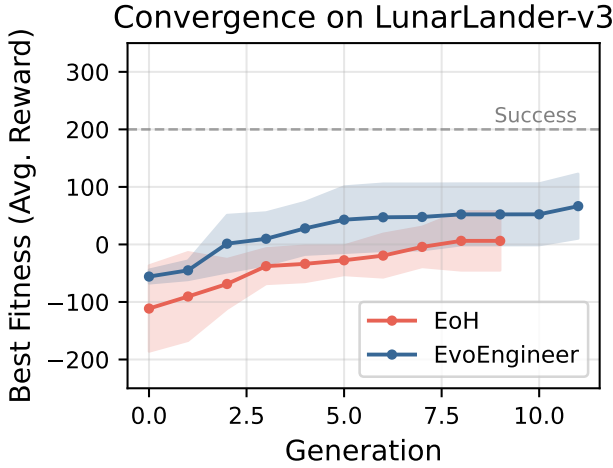


Fig. 1. Best fitness vs. generation for EoH and EvoEngineer. Shaded regions indicate standard deviation across 5 seeds.

higher average reward (214 vs 144) due to more consistent episode performance. This trade-off reflects a fundamental difference: PPO optimizes for expected return through gradient descent, while code evolution explores discrete policy structures that may excel in some episodes but fail in others.

However, this environment-step comparison has important limitations. In simulated environments, environment interactions are computationally cheap; the true cost of LLM-driven evolution lies in API calls to large language models, not in running episodes. With 200 LLM calls at current pricing, EvoEngineer+ incurs non-trivial inference costs, whereas PPO’s 1M environment steps require only local GPU computation. This distinction becomes less significant as LLM inference costs continue to decrease, and reverses entirely in real-world robotics where physical environment interactions are expensive and potentially dangerous.

D. Evolved Policy Analysis

We analyze the best evolved policy (from EvoEngineer+, reward 143.6, 70% success rate) to understand its control strategy. The policy implements a three-phase altitude-based approach, as shown in Figure 2. The evolved code exhibits clear interpretable patterns: phase-based control with different logic for high, mid, and low altitudes; explicit human-readable thresholds for state conditions; and direct mapping from state observations to discrete actions. Unlike neural network weights, every decision boundary can be inspected and modified by human engineers.

VI. DISCUSSION

A. The Role of Context in LLM-Driven Evolution

Our results highlight the critical importance of providing rich, structured context to guide LLM-based code evolution. FunSearch and EoH, despite using sophisticated evolutionary operators, struggle to discover effective control strategies because they lack domain-specific feedback. The LLM generates

```
def policy(state):
    x, y, vx, vy, angle, w, left_leg, right_leg = state

    # Phase 1: High altitude (y > 0.6) - control descent
    if y > 0.6:
        if vy < -1.0: return 2          # main engine: brake
        if angle > 0.05 or w > 0.1:    # right engine:
            return 3                    # correct tilt
        if angle < -0.05 or w < -0.1: # left engine: correct
            return 1                    # tilt

    # Phase 2: Mid altitude (0.2 < y <= 0.6) - correct drift
    elif y > 0.2:
        if vy < -0.5: return 2          # slow descent
        if x > 0.1 and vx > 0.15:      # correct rightward
            return 3                    # drift
        if x < -0.1 and vx < -0.15:   # correct leftward
            return 1                    # drift
        if angle > 0.05: return 3       # attitude correction
        if angle < -0.05: return 1

    # Phase 3: Low altitude (y <= 0.2) - soft landing
    else:
        if vy < -0.2: return 2          # ensure soft
            touchdown
        if x > 0.1: return 3            # fine position
            adjustment
        if x < -0.1: return 1

    return 0 # do nothing: conserve fuel
```

Fig. 2. Best evolved policy (simplified excerpt). The full policy contains additional edge-case handling for angular velocity and leg contact states.

syntactically valid code but cannot reason about *why* a policy fails or *how* to improve it without explicit signals.

EvoEngineer addresses this gap by feeding episode statistics, failure mode analysis, and domain hints back to the LLM. When the model observes that crashes correlate with high descent velocity at low altitude, it can hypothesize that earlier braking is needed and modify thresholds accordingly. This closed-loop feedback transforms the LLM from a blind code mutator into an informed policy designer. The $7\times$ performance gap between EvoEngineer and simpler approaches (66.6 vs. ~ 8 average reward) underscores that context, not just evolutionary structure, drives effective policy synthesis.

B. When to Choose Evolved Code over Neural Networks

The comparison between EvoEngineer+ and PPO reveals a nuanced trade-off. PPO achieves higher average reward (214 vs. 144) because gradient-based optimization produces smooth policies that gracefully handle edge cases. However, EvoEngineer+ achieves a higher success rate (70% vs. 60%) because discrete decision boundaries can execute precise maneuvers when conditions are met, even if they fail abruptly otherwise.

This distinction matters in practice. For applications where task completion is binary (landed or crashed, package delivered or lost, patient stabilized or not), success rate may be the more relevant metric. For applications requiring consistent performance across all episodes, average reward better captures policy quality. Evolved code policies are particularly attractive when human oversight is required: a 59-line Python

function can be reviewed, audited, and certified in ways that a 10K-parameter neural network cannot. Domain experts can directly inspect decision boundaries, verify safety constraints, and modify thresholds without retraining.

C. Scaling and Practical Considerations

Scaling from 45 to 200 LLM calls improves reward from 66.6 to 143.6 and success rate from 40% to 70%, while simultaneously reducing code length from 80 to 59 lines. This pattern suggests that early iterations capture coarse-grained improvements, while later iterations refine edge cases and eliminate redundant logic. Notably, extended evolution produces not only better-performing policies but also simpler ones, indicating that the LLM learns to distill control knowledge into more elegant representations.

The cost structure differs fundamentally from neural network training. LLM API calls dominate the budget for code evolution, while environment interactions are essentially free. For PPO, the situation reverses: GPU computation is the primary cost, and simulation steps are cheap. This makes code evolution particularly attractive for domains where environment interaction is expensive or dangerous, such as physical robotics, where each real-world episode carries time, wear, and safety costs that dwarf API expenses.

D. Limitations

Several limitations constrain the current work. Our evaluation focuses on a single benchmark (LunarLander-v3) with discrete actions; generalization to continuous control, higher-dimensional state spaces, and more complex tasks remains to be demonstrated. The approach inherits the LLM’s biases and capabilities, meaning performance may vary across models and could degrade for domains outside the LLM’s training distribution. Evolved policies exhibit higher variance than neural networks, succeeding decisively or failing completely rather than achieving consistent moderate performance. Finally, while LLM inference costs continue to decrease, extended evolution (200+ calls) still incurs non-trivial API expenses compared to local GPU training.

VII. CONCLUSION

We presented a systematic study of LLM-driven code evolution for control policy synthesis. By comparing three evolution strategies on the LunarLander benchmark, we demonstrated that rich contextual feedback (EvoEngineer) dramatically outperforms minimal-prompt approaches (FunSearch, EoH), achieving $7\times$ higher reward with the same LLM budget. Extended evolution (EvoEngineer+) produces interpretable 59-line policies that achieve 70% success rate, exceeding PPO’s 60% despite lower average reward. The key insight is that LLMs can synthesize effective control logic when provided with structured feedback about behavior, not just fitness scores.

Future work includes extending to continuous action spaces, evaluating on complex locomotion and manipulation tasks, exploring hybrid approaches that combine LLM evolution with gradient-based fine-tuning, and investigating multi-objective

evolution that jointly optimizes performance, interpretability, and robustness.

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Enhancing Trust Through Standards: A Comparative Risk-Impact Framework for Aligning ISO AI Standards with Global Ethical and Regulatory Contexts

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Abstract—As artificial intelligence (AI) reshapes industries and societies, ensuring its trustworthiness—through mitigating ethical risks like bias, opacity, and accountability deficits—remains a global challenge. International Organization for Standardization (ISO) AI standards, such as ISO/IEC 24027 and 24368, aim to foster responsible development by embedding fairness, transparency, and risk management into AI systems. However, their effectiveness varies across diverse regulatory landscapes, from the EU’s risk-based AI Act to China’s stability-focused measures and the U.S.’s fragmented state-led initiatives. This paper introduces a novel Comparative Risk-Impact Assessment Framework to evaluate how well ISO standards address ethical risks within these contexts, proposing enhancements to strengthen their global applicability. By mapping ISO standards to the EU AI Act and surveying regulatory frameworks in ten regions—including the UK, Canada, India, Japan, Singapore, South Korea, and Brazil—we establish a baseline for ethical alignment. The framework, applied to case studies in the EU, US-Colorado, and China, reveals gaps: voluntary ISO standards falter in enforcement (e.g., Colorado) and undervalue region-specific risks like privacy (China). We recommend mandatory risk audits, region-specific annexes, and a privacy-focused module to enhance ISO’s adaptability. This approach not only synthesizes global trends but also offers a replicable tool for aligning standardization with ethical imperatives, fostering interoperability and trust in AI worldwide. Policymakers and standards bodies can leverage these insights to evolve AI governance, ensuring it meets diverse societal needs as the technology advances.

Keywords— AI Standards, Trustworthiness, Regulatory frameworks, Risk impact, AI Ethics

I. INTRODUCTION

A. Background and Context

Artificial intelligence (AI) has emerged as a transformative force, reshaping industries such as healthcare, finance, manufacturing, and education through capabilities like autonomous decision-making and predictive analytics. Its integration into critical societal functions offers immense potential for innovation and productivity, yet it also introduces significant ethical, legal, and societal challenges. Issues such as algorithmic bias, lack of transparency, and accountability deficits have been widely documented, particularly in high-stakes domains like law enforcement and healthcare [1]. For instance, biased AI systems in hiring have disproportionately excluded marginalized groups, raising concerns about fairness and equity [2]. As AI’s societal footprint grows, ensuring its trustworthiness—defined as reliability, safety, and ethical alignment—becomes paramount [3]. Without robust governance, the risks of harm, including privacy violations and unintended consequences, threaten public trust and adoption [4]. These challenges underscore the urgency of

developing frameworks that balance technological advancement with societal well-being, a task increasingly addressed through standardization and regulation.

B. The Role of Standards in AI

Standards play a pivotal role in mitigating AI’s risks by providing a shared framework for consistency, interoperability, and ethical accountability across industries [5]. They serve as a bridge between governance and technical innovation [6] on one side and societal expectations on the other, embedding principles like transparency, fairness, and safety into AI systems [7]. The International Organization for Standardization (ISO), through its ISO/IEC JTC 1/SC 42 committee [8], has developed guidelines addressing data quality, bias, and risk management, aiming to foster trustworthy AI. For example, ISO/IEC 24027 [9] offers tools to assess and reduce unwanted bias, directly tackling ethical concerns. Beyond technical specifications, standards signal a commitment to ethical values, enabling developers to align AI with human rights and regulatory requirements. However, their effectiveness depends on adoption and adaptability to diverse global contexts, where regulatory priorities—such as the EU’s risk-based approach or China’s focus on social stability—differ significantly [10]. This variability highlights the need for a nuanced evaluation of standards’ impact on ethical risks.

C. Need for a New Analytical Approach

While ISO standards provide a universal foundation for responsible AI, their application across heterogeneous regulatory regimes reveals limitations. Existing mappings, such as those aligning ISO standards with the EU AI Act, focus narrowly on compliance within specific frameworks, overlooking gaps in addressing region-specific ethical risks [11]. For instance, the voluntary nature of ISO/IEC 23894 on risk management [12] may suffice in the EU’s structured environment but falters in the U.S., where state-level mandates like Colorado’s AI Act [13] contrast with federal deregulation [14]. Similarly, China’s emphasis on social harmony requires privacy safeguards beyond ISO’s current scope. These disparities suggest that a static, one-size-fits-all approach to standardization is insufficient. This paper proposes a Comparative Risk-Impact Assessment Framework to address this gap, offering a systematic method to evaluate and enhance ISO standards’ effectiveness in mitigating ethical risks across diverse global contexts, thereby advancing their role in building trustworthy AI.

D. Scope and Objectives of the Paper

This paper aims to enhance trust in AI systems by synthesizing global regulatory trends and ISO standardization

efforts while introducing an original contribution: the Comparative Risk-Impact Assessment Framework. Beyond mapping ISO standards to the EU AI Act, we develop and apply this framework to assess how well these standards address ethical risks—such as bias, transparency, and accountability—in varying regulatory landscapes, including the EU, U.S. states, China, and other key regions. By identifying gaps and proposing targeted enhancements to ISO standards, we seek to strengthen their adaptability and impact, fostering responsible AI development worldwide. The scope encompasses an overview of ISO’s role, a survey of emerging AI regulations, a detailed standards mapping, and the framework’s application to case studies, culminating in actionable recommendations. This approach bridges theoretical standardization with practical governance, offering a novel tool for aligning AI innovation with ethical and societal imperatives.

II. THE IMPORTANCE OF AI STANDARDIZATION

A. Promoting Interoperability and Ethical Integration

AI standardization fosters interoperability by establishing a common technical and ethical language, enabling seamless collaboration across systems and stakeholders [5]. This consistency is critical as AI applications—like autonomous vehicles or medical diagnostics—span borders and industries, requiring compatible frameworks to ensure functionality and safety [7]. Beyond technical integration, standards embed ethical principles such as fairness and inclusivity into AI design, reducing risks of harm to vulnerable populations [15]. For instance, interoperable standards can prevent fragmented AI ecosystems that exacerbate inequities, as seen in biased facial recognition systems disproportionately misidentifying minorities [16]. By harmonizing technical and ethical dimensions, standardization lays the groundwork for innovation that aligns with societal values, a foundation this paper builds upon to assess global regulatory alignment.

B. Enhancing Trust through Ethical Safeguards

Standards enhance trust by providing ethical safeguards that mitigate AI’s inherent risks, such as bias, discrimination, and opacity [17]. Defined protocols—like ISO/IEC 24027 [9] for bias assessment—offer measurable criteria to ensure reliability and fairness, addressing public concerns about “black box” AI systems [18]. Transparency requirements, a cornerstone of many standards, compel developers to disclose decision-making processes, building confidence among users and regulators [4]. Research shows that trust in AI increases when ethical safeguards are visible and enforceable, as evidenced by public backlash to unregulated AI in hiring [2]. These safeguards are vital for widespread adoption, yet their effectiveness varies across regulatory contexts, necessitating a comparative approach to evaluate their impact.

C. Supporting Ethical and Responsible AI Development

Standardization supports ethical and responsible AI development by embedding societal values into technical processes, addressing concerns like privacy and accountability [19]. Standards mandate practices that reduce ethical risks—for example, ISO/IEC 5259 [20] ensures data quality to prevent biased outcomes in machine learning. This alignment with human rights and fairness is critical in sensitive applications, such as healthcare AI, where errors can perpetuate harm [21]. By providing a structured framework, standards guide developers toward accountability, ensuring AI serves the public good rather than amplifying existing inequalities [22]. However, the diversity of global ethical

priorities highlights the need for adaptive standards, a challenge this paper tackles through its proposed framework.

D. Facilitating Global Harmonization

AI standardization facilitates global harmonization by creating consistent guidelines that transcend national boundaries, fostering equitable access and collaboration [5]. International standards, such as those from ISO/IEC JTC 1/SC 42, reduce trade barriers and align AI practices with universal ethical norms, leveling the playing field for innovation [10]. This is particularly relevant as countries like the EU and China pursue divergent regulatory paths—risk-based versus control-focused—yet share common goals of fairness and safety [14]. Harmonization ensures that ethical disparities, such as unequal privacy protections, are minimized, benefiting global society. Yet, achieving true harmonization requires standards to adapt to regional nuances, a gap this paper’s framework seeks to address.

E. Guiding Ethical Innovation and Regulatory Compliance

Standards guide ethical innovation by identifying benchmarks for improvement, such as fairness and inclusivity, while ensuring compliance with evolving regulations [7]. They provide evaluation criteria—like ISO/IEC 25059’s [23] quality model—that help organizations meet legal and ethical mandates, reducing risks of misuse. As regulatory frameworks, such as the EU AI Act, impose stringent requirements on high-risk AI, standards serve as a compliance roadmap, bridging innovation and accountability [11]. However, their voluntary nature can limit enforceability in less regulated contexts, like the U.S., where state laws outpace federal action [13]. This tension underscores the need for a comparative risk-impact analysis to enhance standards’ global applicability, a core contribution of this work.

III. OVERVIEW OF ISO AND ITS ROLE IN AI STANDARDIZATION

A. Introduction to ISO

The International Organization for Standardization (ISO), founded in 1947, is a global body dedicated to developing voluntary, consensus-based standards that enhance quality, safety, and efficiency across industries. Initially focused on mechanical engineering—its first standard, ISO/R 1:1951, standardized industrial length measurements—ISO’s scope expanded over decades to address diverse domains, from shipping container standardization in the 1960s to environmental management with ISO 14000 in the 1990s [24]. Today, with 169 member countries and over 24,000 standards, ISO facilitates international trade, reduces technical barriers, and supports regulatory compliance. Its evolution into technology sectors, including artificial intelligence (AI), reflects a response to emerging global challenges, such as cybersecurity and ethical AI governance [25]. ISO’s collaborative, market-driven approach ensures standards reflect stakeholder needs, making it a cornerstone for aligning AI innovation with societal values, a role this paper leverages to assess ethical risk mitigation.

B. ISO’s Approach to Standard Development in Technology

ISO’s approach to developing technology standards is characterized by collaboration, market relevance, and technical rigor, ensuring adaptability to complex fields like AI [5]. It convenes experts from industry, academia, government, and consumer groups worldwide, fostering diverse input through technical committees. For instance, proposals are market-driven, responding to technological trends and societal needs, as seen in the rapid development of

AI standards post-2018. The process involves rigorous drafting, public consultation, and iterative revisions, ensuring standards like ISO/IEC 24028 [26] on AI trustworthiness are technically sound and widely applicable [27]. Regular updates maintain relevance, with committees like JTC 1 adapting to AI's evolving risks [28]. This methodical approach underpins ISO's ability to guide ethical AI development, though its voluntary nature poses challenges in enforcement, a limitation this paper's framework seeks to address across regulatory contexts.

C. ISO/IEC JTC 1/SC 42 Committee

The ISO/IEC JTC 1/SC 42 [8] committee, established in 2017 under the Joint Technical Committee 1 (JTC 1) of ISO and the International Electrotechnical Commission (IEC), is the primary body for AI standardization (ISO/IEC, 2023). SC 42 addresses the AI ecosystem through specialized working groups: WG1 (foundational standards), WG2 (data and analytics), WG3 (trustworthiness), WG4 (use cases), and WG5 (computational approaches), alongside joint working groups like JWG1 (governance) with other subcommittees (ISO/IEC, 2023). It has produced key standards, such as ISO/IEC 23894 [12] for risk management and ISO/IEC 24368 [29] for ethical considerations, aligning AI with global norms [18]. SC 42 collaborates with external bodies, like IEEE and OECD, to ensure interoperability and ethical coherence [5]. As of 2025, its focus on trustworthiness and governance provides a robust baseline for this paper's Comparative Risk-Impact Assessment Framework, which evaluates these standards' effectiveness across diverse regulatory landscapes.

IV. GLOBAL REGULATORY FRAMEWORKS

A. Emerging AI Regulations

The rapid proliferation of artificial intelligence (AI) has prompted diverse regulatory responses worldwide, each addressing distinct ethical risks like bias, transparency, and societal harm. This subsection surveys frameworks in ten key regions, setting the stage for the Comparative Risk-Impact Assessment Framework by highlighting varied approaches to ethical governance.

The United Kingdom pursues a "pro-innovation" approach, rooted in its 2023 AI Regulation White Paper [30], favoring a principles-based, sector-specific framework over broad legislation. Regulators like the Financial Conduct Authority (FCA) enforce ethical oversight, with the Consumer Duty (effective 2023) tackling fairness in AI-driven finance (FCA, 2023). The Information Commissioner's Office (ICO) ensures transparency via AI auditing [31]. The 2023 AI Safety Summit and the AI Safety Institute underscore global leadership, though the Labour government's 2024 consultation signals a shift toward mandatory high-risk AI rules.

The United States exhibits a fragmented approach, with federal deregulation under Trump's 2025 Removing Barriers to American Leadership in AI executive order prioritizing innovation over ethical safeguards [32]. State-level efforts contrast sharply: Colorado's AI Act targets bias in high-risk systems [13], Illinois' HB 3773 (2024) [33] addresses employment discrimination, and California's SB-942 (2026) mandates transparency for generative AI [13]. Tennessee's ELVIS Act (2024) [34] mitigates deepfake risks, highlighting varied ethical priorities [2].

Since launching its *New Generation AI Development Plan* [35] in 2017, China has aimed to balance AI innovation with regulatory control. It has introduced key frameworks such as

the *Interim Measures for Generative AI Services* [36] and the 2023 *Deep Synthesis Measures* to address algorithmic accountability and deepfake risks [10]. In 2024, the country released *Ethical Norms for AI* that emphasize social stability. This commitment was further reflected in its participation at the 2023 UK AI Safety Summit.

Canada has released Pan-Canadian AI Strategy [37], by which, through the Standards Council of Canada, the Government of Canada is supporting efforts to advance the development and adoption of standards related to AI. The Artificial Intelligence and Data Act (AIDA) [38] which is under review, targets accountability in high-impact AI, while the Personal Information Protection and Electronic Documents Act (PIPEDA) [39] ensures privacy.

Australia integrates ethics via the 2019 National AI Ethics Framework [40], emphasizing transparency across eight principles, supported by the 2021 AI Action Plan [41]. The ACCC and OAIC enforce consumer and privacy protections, with 2024 consultations proposing mandatory high-risk AI rules [19].

India advances ethical AI with the 2018 #AIforAll Strategy, targeting inclusivity in healthcare and education, guided by the 2021 Responsible AI Principles (NITI Aayog, 2021) [42]. The 2024 IndiaAI Mission [43] and forthcoming Digital India Act address bias in high-risk systems, bolstered by GPAI leadership [44].

Japan prioritizes societal harmony via the 2019 Social Principles of Human-Centric AI [45], focusing on safety and fairness, updated in the 2024 AI Strategy (METI, 2024). The 2024 AI Guidelines for Business [46] and Hiroshima AI Process [47] reflect a soft-law approach with global resonance [18].

Singapore leads with the 2019 Model AI Governance Framework [48], updated 2024 [49], tackling transparency in AI deployment, complemented by the 2024 Generative AI Evaluation Sandbox. The National AI Strategy 2.0 (2023) [50] aligns ethics with innovation.

The Ministry of Science and ICT (MSIT) of South Korea announced the passage of the Basic Act on Artificial Intelligence (AI Basic Act) [51] by the National Assembly on December 26, 2024, set to take effect in January 2026. This landmark legislation, making Korea the second country globally to enact such a law, aims to bolster national AI competitiveness and establish a trustworthy AI framework amid growing global competition, following models like the EU AI Act (2024). Developed over four years through bipartisan efforts, the Act outlines a national AI governance structure, including a triennial AI Master Plan, the National AI Committee chaired by the President, and an AI Safety Institute. It supports the AI industry through R&D, data training, AI clusters, and talent development, while addressing risks from high-risk and generative AI with transparency, safety, and operator accountability measures, positioning Korea to become a top global AI leader.

Brazil has made steady progress in shaping its AI governance landscape. Initially, Bill 21/2020, introduced in 2020 and approved by the House of Representatives in 2021, set forth a proposed legal framework for artificial intelligence, emphasizing transparency and accountability in high-risk systems. This framework evolved through legislative deliberations and was enacted as the Brazil AI Act in 2024 [52], incorporating guidelines for ethical deployment and oversight. Complementing this, Brazil's National AI Strategy 2021 [53], places a strong emphasis on inclusivity

and fairness, aligning with global efforts to address algorithmic bias, as highlighted by Obermeyer et al. [21].

B. The European Union’s AI Act

The European Union Artificial Intelligence Act (AIA) [54], effective from August 2024, establishes a comprehensive, risk-based framework for ethical AI. This framework was informed by earlier analyses, such as Veale and Borgesius’s 2021 study [14] of the draft AIA, which provided insights into its proposed scope, including its application to providers and deployers within and beyond the EU. The AIA bans unacceptable-risk systems like mass surveillance and imposes strict requirements on high-risk AI systems. For minimal-risk systems, it mandates transparency, a principle that resonates with the ethics-based auditing approach outlined by Mökander et al [11]. General-purpose AI systems face additional obligations if they pose systemic risks. Supported by regulatory sandboxes and national enforcement, the AIA integrates with existing laws to set a global benchmark for ethical AI governance. This paper uses the AIA as a reference to evaluate the alignment of ISO standards with these ethical principles.

V. MAPPING OF ISO STANDARDS

A. Data and Governance-Related Standards

Data quality and governance are essential for building trustworthy AI, a priority highlighted by the EU AI Act’s demand for reliable datasets in high-risk systems. ISO/IEC 5259-1:2021 offers a framework for managing data quality in analytics and machine learning, focusing on accuracy, completeness, and consistency to prevent errors that might cause biased results. For example, it supports the EU’s requirement for representative training data to lower the risk of discrimination. Additionally, ISO/IEC 38507:2022 [55] provides governance guidelines for organizations using AI, aligning with the EU’s accountability rules by clarifying oversight roles and responsibilities. While these standards promote ethical data practices, they aren’t mandatory, which poses challenges in regions with less strict regulations [11]. This connection helps the framework assess how data-related ethical risks vary across different global contexts.

B. Ethics, Fairness, and Transparency-Related Standards

The EU AI Act, requires high-risk AI systems to be transparent and fair. These goals align with ISO standards that focus on ethical AI practices. For example, ISO/IEC 24368:2022 [29] sets out principles like fairness, transparency, and respect for human values, which support the EU’s push for explainable AI and reducing bias. Likewise, ISO/IEC 24027:2021 offers ways to spot and fix unwanted bias in AI, addressing issues like those found in healthcare algorithms. While these standards fit well with EU rules—such as the need to label AI-generated content—their voluntary nature means they’re not always enforced outside strict regulatory environments. This gap is a central concern for the Comparative Risk-Impact Assessment Framework, which examines how effectively these standards promote fairness and transparency worldwide..

C. Risk Management-Related Standards

Risk management is central to the EU AI Act’s classification of AI systems by risk level, requiring robust mitigation for high-risk applications. ISO/IEC 23894:2023 provides a comprehensive risk management framework for AI, guiding organizations in identifying, assessing, and

treating risks like safety failures or privacy breaches (ISO/IEC 23894:2023). It aligns with the EU’s human oversight and documentation mandates, providing a structured approach to compliance. Additionally, ISO/IEC 24028:2020 focuses on trustworthiness in AI through resilience and security, which is supported by research on Trustworthy AI [3]. This research emphasizes principles like beneficence and non-maleficence, essential for managing systemic risks in general-purpose AI as per the EU’s provisions. While these standards are effective in structured regulatory environments like that of the EU, their flexibility may be insufficient in regions prioritizing different risks, such as social stability, a challenge addressed by our framework.

D. Quality Management-Related Standards

Quality management ensures AI systems meet performance and ethical benchmarks, a priority under the EU AI Act’s technical robustness requirements. ISO/IEC 25059:2023 establishes a quality model for AI-based systems, defining criteria like reliability and usability, which align with the EU’s emphasis on system accuracy and safety. This standard supports compliance by offering evaluation metrics for high-risk AI, such as autonomous vehicles [7]. Paired with ISO/IEC 5338:2023, which provides lifecycle management for AI development, it ensures consistent quality from design to deployment. Yet, their emphasis on technical quality might overlook societal impacts like equity—an issue Mittelstadt flagged [19] in his critique of principle-based AI ethics. The Comparative Risk-Impact Assessment Framework explores this gap across global regulatory contexts.

VI. THE COMPARATIVE RISK-IMPACT ASSESSMENT FRAMEWORK

A. Framework Description

The Comparative Risk-Impact Assessment Framework is a novel tool designed to evaluate and enhance the effectiveness of ISO AI standards in mitigating ethical risks across diverse global regulatory contexts. It addresses the limitation of static standard mappings by systematically analyzing how well ISO standards align with region-specific ethical priorities, offering a prescriptive approach to strengthen trustworthiness in AI systems [11]. The framework comprises three components:

Risk Identification: Identifies key ethical risks—bias, transparency, accountability, privacy, and societal harm—drawn from ISO/IEC 24368 and global regulations (ISO/IEC, 2023a; [44].

Impact Scoring: Assesses each risk’s severity (1-5) and likelihood (1-5) within a regulatory context, producing a composite impact score (severity $\times$ likelihood, max 25), reflecting local ethical and legal priorities [3].

Standards Alignment: Scores ISO standards’ mitigation effectiveness (1-5) for each risk, based on their scope, specificity, and enforceability [18].

For example, bias in hiring AI might score high impact in a region with anti-discrimination laws but lower where unregulated, while ISO/IEC 24027’s voluntary bias tools might score moderately due to lack of mandates. This structured approach, visualized in Table 1 (below), enables gap analysis and tailored enhancements, advancing ISO’s role in ethical AI governance beyond current mappings [14]

TABLE I. FRAMEWORK COMPONENTS EXAMPLE

<i>Risk</i>	<i>Severity</i>	<i>Likelihood</i>	<i>Impact Score</i>	<i>ISO Standard</i>	<i>Mitigation Score</i>
Bias	4	3	12	24027	3
Transparency	3	4	12	24368	4

B. Application to Case Studies

The framework is applied to three regulatory contexts to demonstrate its utility: the EU (EU AI Act), US-Colorado (Colorado AI Act), and China (Generative AI Measures).

European Union (EU AI Act): The EU’s risk-based approach prioritizes bias and transparency in high-risk AI (European Commission, 2024). Bias scores high (Severity 4, Likelihood 4, Impact 16) due to strict anti-discrimination rules, with ISO/IEC 24027 scoring 4 for its robust bias assessment tools, aligning well with EU mandates [21]. Transparency (Severity 3, Likelihood 5, Impact 15) reflects disclosure requirements, matched by ISO/IEC 24368’s explainability focus (Score 4) [18]. Accountability scores moderately (Impact 12), with ISO/IEC 23894 (Score 3) lacking enforceable oversight mechanisms [11].

United States-Colorado (Colorado AI Act): Colorado targets algorithmic discrimination in hiring and healthcare [13]. Bias is critical (Severity 5, Likelihood 3, Impact 15) due to legal mandates, but ISO/IEC 24027 scores only 2, as its voluntary nature limits enforcement in a deregulated federal context [2]. Transparency (Severity 3, Likelihood 3, Impact 9) is less emphasized, with ISO/IEC 24368 (Score 3) offering guidance but no binding rules. Accountability (Impact 10) aligns poorly with ISO/IEC 38507 (Score 2), highlighting governance gaps.

China (Generative AI Measures): China prioritizes social stability and privacy in AI content [10]. Privacy scores high (Severity 4, Likelihood 4, Impact 16) due to state oversight, but ISO/IEC 24028 (Score 2) lacks specific privacy controls, missing China’s focus. Transparency (Severity 2, Likelihood 3, Impact 6) is secondary, with ISO/IEC 24368 (Score 3) partially relevant. Societal harm (Impact 18) is critical, yet ISO/IEC 23894 (Score 3) offers generic risk management, not tailored to stability concerns.

C. Findings and Gap Analysis

The framework reveals significant gaps in ISO standards’ adaptability. In the EU, high alignment exists for bias and transparency (Scores 4), but accountability lags due to voluntary adoption (Score 3) (European Commission, 2024). Colorado shows weaker alignment, with bias mitigation (Score 2) and accountability (Score 2) undermined by ISO’s lack of enforceability, exacerbating risks in a fragmented U.S. landscape [13]. China highlights a privacy gap (Score 2) and insufficient focus on societal harm (Score 3), reflecting ISO’s Western-centric design. Across cases, voluntary standards struggle where regulations demand binding rules [19], and regional ethical priorities (e.g., stability in China) are underrepresented, limiting ISO’s global effectiveness.

D. Recommendations for Enhancing ISO Standards

To address these gaps, we recommend some enhancements to ISO standards. Building on the ethical framework for a good AI society proposed by Floridi et al. [4] which emphasizes principles like data governance and transparency, and the trustworthy AI concept outlined by Thiebes et al. [3], focusing on beneficence and justice, we propose the following seven enhancements to ISO standards. These recommendations aim to mitigate ethical risks,

enhance trust, and ensure interoperability worldwide by establishing foundational technical standards and integrating them into existing ISO frameworks:

- **Development of an AI-Augmented Cybersecurity Standard:** Create a new ISO standard focused on securing AI systems against adversarial threats. This standard will define technical frameworks, including methodologies for robustness testing against adversarial attacks, secure model deployment protocols with end-to-end encryption, and AI-driven threat detection mechanisms, to ensure AI systems are resilient and trustworthy.
- **Development of an AI Edge Computing Standard:** Establish a new ISO standard for AI in edge computing and federated learning systems. This standard will provide compatibility frameworks, including interoperability protocols for edge AI models, privacy-preserving machine learning techniques for decentralized training, and guidelines for secure operation in distributed environments, addressing efficiency and privacy needs.
- **Mandatory Risk Audit Standard:** Develop a new ISO standard mandating periodic risk audits for high-impact AI, incorporating the technical frameworks from the AI-Augmented Cybersecurity Standard (Recommendation 1) for robustness testing and secure deployment, to enhance accountability beyond the voluntary framework of ISO/IEC 23894 (e.g., addressing needs in the EU and Colorado).
- **Region-Specific Annexes:** Add annexes to standards like ISO/IEC 24368 and 24028, tailoring guidance [sic] guidance to regional risks—e.g., privacy for China, bias for Colorado—improving adaptability, supported by the AI Edge Computing Standard’s (Recommendation 2) privacy-preserving techniques for region-specific compliance.
- **Strengthened Privacy Module:** Expand ISO/IEC 24028 with a privacy-specific module, aligning with global data protection trends (e.g., China’s needs), reinforced by the AI Edge Computing Standard’s (Recommendation 2) federated learning protocols and the AI-Augmented Cybersecurity Standard’s (Recommendation 1) secure model training guidelines to ensure robust privacy protections.
- **Integration of AI-Augmented Cybersecurity Frameworks:** Incorporate the AI-Augmented Cybersecurity Standard (Recommendation 1) into existing ISO frameworks, applying its standardized methodologies for robustness testing, secure deployment, and AI-driven threat detection to complement the risk audit standard and enhance overall system trustworthiness.
- **Integration of AI Edge Computing Frameworks:** Integrate the AI Edge Computing Standard (Recommendation 2) into ISO frameworks, utilizing its compatibility protocols and privacy-preserving techniques to support the privacy module and ensure secure, efficient AI operation in distributed environments.

These recommendations ensure ISO standards better mitigate ethical risks, enhancing trust and interoperability worldwide [3].

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